STATS 280 - Midterm 2

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Q.1 (30)

Q.1 a

(5 points) This is the chance that the card is a spade, which is $\frac{13}{52} = \frac{1}{4}$.

Q.1 b

(5 points) This is the chance that the first card is not a spade and the second card is a spade, which is $\frac{39}{52}\frac{13}{51} = 0.19$.

Q.1 c

(5 points) This is the chance that the first two cards are not spades and the third card is a spade, which is $\frac{39}{52}\frac{38}{51}\frac{13}{50} = 0.15$.

Q.1 d

(7 points) You will draw more than five cards precisely when none of the first five cards are spades, and this chance is $\frac{39}{52}\frac{38}{51}\frac{37}{50}\frac{36}{49}\frac{35}{48} = 0.22$.

Q.1 e

(8 points) The chance of five or fewer cards is $1 - 0.22 = 0.78$ from part d). The chance of six or fewer cards is $1 - \frac{39}{52}\frac{38}{51}\frac{37}{50}\frac{36}{49}\frac{35}{48}\frac{34}{47} = 0.84$, seven or fewer cards is $1 - \frac{39}{52}\frac{38}{51}\frac{37}{50}\frac{36}{49}\frac{35}{48}\frac{34}{47}\frac{33}{46} = 0.89$, and eight or fewer cards is $1 - \frac{39}{52}\frac{38}{51}\frac{37}{50}\frac{36}{49}\frac{35}{48}\frac{34}{47}\frac{33}{46}\frac{32}{45} = 0.92$. Hence eight is the smallest number of cards with the chance of at least 90% of drawing that many cards or fewer.

Q.2 (30)

Q.2 a

(8 points) The maximum possible percentage is $0.49 + 0.21 - 0 = 0.70$, which would happen when no out of state students are first year students.

Q.2 b

(7 points) This is $0.49 + 0.21 - 0.12 = 0.58$.

Q.2 c

(8 points) If being out of state and being a first year student were independent, then the chance of being out of state and a first year student would be $0.49 * 0.21 = 0.10$, and not 0.12. Hence being out of state and being a first year student are not independent.

Q.2 d

(7 points) This could only happen if all students with the characteristic are out of state students. This follows because the percentage of students who are out of state OS or have the characteristic C is $0.49 + P(C) - P(C \text{ and } OS) = 0.49$, so that $P(C) = P(C \text{ and } OS)$.

Q.3 (30)

The probability that the butterfly is from species A is (10 points):

$$\begin{aligned} P(A|wing) &= P(wing|A)P(A)/P(wing) && (1)\\ &= \frac{0.6 * 0.3}{0.6 * 0.3 + 0.2 * 0.5 + 0.9 * 0.2} && (2)\\ &= 0.39, && (3) \end{aligned}$$

the probability that the butterfly is from species B is (10 points):

$$\begin{aligned} P(B|wing) &= P(wing|B)P(B)/P(wing) && (4)\\ &= \frac{0.2 * 0.5}{0.6 * 0.3 + 0.2 * 0.5 + 0.9 * 0.2} && (5)\\ &= 0.22 && (6) \end{aligned}$$

Q.3c

(10 points) and the probability that the butterfly is from species C is (10 points):

$$\begin{aligned} P(C|wing) &= P(wing|C)P(C)/P(wing) && (7)\\ &= \frac{0.9 * 0.2}{0.6 * 0.3 + 0.2 * 0.5 + 0.9 * 0.2} && (8)\\ &= 0.39 && (9) \end{aligned}$$

Q.4 (40)

Q.4a

(8 points) This will be the hypergeometric probability $\binom{5}{3}\binom{10}{7}/\binom{15}{10}$. Simplifying gives $\binom{5}{3} = \frac{5!}{3!2!} = \frac{5*4}{2} = 10$, $\binom{10}{7} = \frac{10!}{7!3!} = \frac{10*9*8}{3*2} = 120$, $\binom{15}{10} = \frac{15!}{10!5!} = \frac{15*14*13*12*11}{5*4*3*2} = 3003$. Hence the probability is $\frac{10*120}{3003} = 0.4$.

Q.4b

(5 points) The command will be 1 - phyper(4,10,5,10). Alternatively, at least 5 women will correspond to at most 5 men, and the chance of this is 1 since there are only 5 men.

Q.4c

(8 points) This is now a binomial experiment, and so the probability will be $\binom{10}{3}(5/15)^3(10/15)^7$. Simplification gives $\binom{10}{3} = \frac{10!}{3!7!} = \frac{10*9*8}{3*2} = 120$. Hence the probability is $120 * (5/15)^3(10/15)^7 = 0.26$.

Q4.d

(5 points) The command will be pbinom(4,10,1/3).

Q.4e

(9 points) This is the opposite of at most 2 of the volunteers being men. The chance that 0 volunteers are men is $\frac{2}{3}^{10} = 0.017$, the chance that 1 volunteer is a man is $10 * (1/3) * (2/3)^9 = 0.087$, and the chance that 2 volunteers are men is $\binom{10}{2}(1/3)^2(2/3)^8 = \frac{10*9}{2}(1/3)^2(2/3)^8 = 0.195$. Hence the probability is $1 - (0.017 + 0.087 + 0.195) = 0.70$

Q.4f

(5 points) The command will be dbinom(3,10,1/3).

Q.5 (30)

Q5.a

(10 points) One solution among many:

```
cointoss = function(p) {
    val = runif(1)
    if (val <= p) {return(1)}
    else {return(0)}
    }
```

Q.5 b

(10 points)
Blank 1: n1/(n0+n1)
Blank 2: n0+1
Blank 3: draw==1
Blank 4: i
Blank 5: n1/(n1 + n0 - i)

Q.5 c

(10 points)
Blank 1: i in 1:1000000
Blank 2: i
Blank 3,4: 39,13
Blank 5: mean(cards_drawn)

Q.6 (40)

Q.6 a

(10 points) The expected value is $82 * 0.78 = 63.96$ and the standard error $\sqrt{0.78 * (1 - 0.78) * 82} = 3.75$

Q.6 b

(5 points) The command is 1-pbinom(59,82,0.78).

Q.6 c

(15 points) Converting to standard units gives

$$z_1 = (59.5 - 63.96)/3.75 = -1.19 \tag{10}$$

The table in the back of the book gives roughly $0.76 + (1 - 0.76)/2 = 0.88$ as the chance of winning 60 or more games.

Q.6 d

(10 points) The expected number of wins after 68 games is $68 * 0.78 = 53.04$ and the standard error is $\sqrt{0.78 * 0.22 * 68} = 3.42$. Converting to standard units for the normal approximation gives $(49.5 - 53.04)/3.42 = -1.04$. The chance of 49 or fewer wins will be the left hand tail, which is approximately $(1 - 0.70)/2 = 0.15$. The Celtics have won fewer games than we would expect, but not enough fewer to say that the assumptions in part a) are not realistic.

7 (15 points extra credit)

We know that $X + Y$ takes on possible values of $0, 1, 2$, and so

$$E(X + Y) = 0 * P(X + Y = 0) + 1 * P(X + Y = 1) + 2 * P(X + Y = 2). \tag{11}$$

We also know that $X + Y = 0$ only when $X = 0, Y = 0$, that $X + Y = 1$ when $X = 0, Y = 1$ or $X = 1, Y = 0$, and $X + Y = 2$ only when $X = 1, Y = 1$. Hence

$$E(X + Y) = 1 * (P(X = 0, Y = 1) + P(X = 1, Y = 0)) + 2 * P(X = 1, Y = 1) \tag{12}$$

since $X = 0, Y = 1$ and $X = 1, Y = 0$ are mutually exclusive. We can split up $2 * P(X = 1, Y = 1) = P(X = 1, Y = 1) + P(X = 1, Y = 1)$ and rewrite the expected value equation as

$$E(X + Y) = (P(X = 0, Y = 1) + P(X = 1, Y = 1)) + (P(X = 1, Y = 0) + P(X = 1, Y = 1)). \tag{13}$$

The term in the first set of parentheses is $P(Y = 1)$ and the term in the second set of parentheses is $P(X = 1)$ by the law of total probability. We also know that $E(X) = 0 * P(X = 0) + 1 * P(X = 1) = P(X = 1)$ and $E(Y) = 0 * P(Y = 0) + 1 * P(Y = 1) = P(Y = 1)$. Hence

$$E(X + Y) = P(X = 1) + P(Y = 1) = E(X) + E(Y). \tag{14}$$